

Use the `ip address` command to find the IPv4 address and hardware address of the local ethernet card interface (Typically beginning with `eth`, `ens`, or `enp`).

- Include both in your lab notebook

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:13:a0:c6 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 131.252.208.103/24 metric 100 brd 131.252.208.255 scope global dynamic ens3
        valid_lft 12540sec preferred_lft 12540sec
```

Perform a `netstat -rn` to list the route table for the machine.

- What is the default router's IP address (e.g. the gateway address for the default route 0.0.0.0/0)

Default Gateway: 131.252.208.1

Perform an `arp` command (both with and without the `-n` flag) on the IP address of the router.

- What is the name of the default router and its hardware address?

Name: router.seas.pdx.edu Hardware Address: 00:00:5e:00:01:01

- How many entries are there in the ARP table?

29 Total

```
arp -a | sort -k ...
```

- List any IP addresses that share the same hardware address

131.252.208.118 and 131.252.208.53 share 52:54:00:30:e3:f2
omr-rdns-01.cat.pdx.edu (131.252.208.118) at 52:54:00:30:e3:f2 [ether] on ens3
rdns.cat.pdx.edu (131.252.208.53) at 52:54:00:30:e3:f2 [ether] on ens3

```
arp -a | sort -k ... | awk '{print $...}' | uniq | wc -l
```

- How many less hardware addresses are there than IP addresses in the ARP table?

There are 28 hardware addresses and 29 IP addresses, therefore one less.

```
arp -an | awk -F '...' '{print ...}' > arp_entries
```

- Include the command in your lab notebook

```
arp -an | awk -F '[]' '{print $2}' > arp_entries
```

- What network prefix do most of the IP addresses in the ARP table share?

Every entry in the arp entries file uses the “131.252.208.xxx” prefix.

```
ip address
```

- Include both in your lab notebook

```
2: ens4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc mq state UP group default qlen 1000
    link/ether 42:01:0a:8a:00:02 brd ff:ff:ff:ff:ff:ff
    inet 10.138.0.2/32 metric 100 scope global dynamic ens4
        valid_lft 64430sec preferred_lft 64430sec
    inet6 fe80::4001:aff:fe8a:2/64 scope link
        valid_lft forever preferred_lft forever
```

IP = 10.138.0.2, MAC = 42:01:0a:8a:00:02

```
netstat -rn
```

- What is the default router's IP address (e.g. the gateway address for the default route 0.0.0.0/0)

The gateway address is 10.138.0.1

```
arp <IP_address_of_router>
```

- What is the default router's hardware address?

42:01:0a:8a:00:01

- Show a screenshot of the output for the scan for your lab notebook.

```
jgafron@course-vm:~$ nmap 10.138.0.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2025-01-09 05:39 UTC
Nmap scan report for course-vm.c.cloud-gafron-jgafron.internal (10.138.0.2)
Host is up (0.00020s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
3389/tcp  open  ms-wbt-server

Nmap scan report for wordpress-1-vm.c.cloud-gafron-jgafron.internal (10.138.0.3)
Host is up (0.00037s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
10000/tcp open  snet-sensor-mgmt

Nmap scan report for appspace-byol-1.c.cloud-gafron-jgafron.internal (10.138.0.4)
Host is up (0.0011s latency).
Not shown: 991 filtered ports
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
808/tcp   open  ccproxy-http
1030/tcp  open  iadl
1801/tcp  open  msmq
2103/tcp  open  zephyr-clt
2105/tcp  open  eklogin
2107/tcp  open  msmq-mgmt
3389/tcp  open  ms-wbt-server

Nmap scan report for django-1-vm.c.cloud-gafron-jgafron.internal (10.138.0.5)
Host is up (0.00068s latency).
Not shown: 999 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
```

- How many subnetworks are created initially on the default network? How many regions does this correspond to? (Use a pipe to pass output to grep in order to return specific lines of output and then another to pass output to wc to count them: `| grep default | wc -l`)

There are 82 subnetworks created initially, which corresponds to 82 regions.

- Given the CIDR prefix associated with each subnetwork, how many hosts does each subnetwork support?

Each network allows for 4024 hosts

- Which CIDR subnetworks are these instances brought up in? Do they correspond to the appropriate region based on the prior commands?

```
NAME: default
REGION: us-west3
NETWORK: default
RANGE: 10.180.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
```

```
NAME: default
REGION: us-west4
NETWORK: default
RANGE: 10.182.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:
```

NAME: instance-2

ZONE: us-west3-b

MACHINE_TYPE: n1-standard-1

PREEMPTIBLE:

INTERNAL_IP: 10.180.0.2

NAME: instance-1

ZONE: us-west4-b

MACHINE_TYPE: n1-standard-1

PREEMPTIBLE:

INTERNAL_IP: 10.182.0.2

EXTERNAL_IP: 34.

Yes! Instance-1 was brought up in the 10.180.0.0/20 subnet and Instance-2 was brought up in the 10.182.0.0/20 subnet. Both of which correspond to the appropriate region.

From instance-1, perform a ping to the Internal IP address of instance-2. Take a screenshot of the output.

```

jgafron@instance-1:~$ ping 10.180.0.2
PING 10.180.0.2 (10.180.0.2) 56(84) bytes of data.
 64 bytes from 10.180.0.2: icmp_seq=1 ttl=64 time=9.66 ms
 64 bytes from 10.180.0.2: icmp_seq=2 ttl=64 time=9.22 ms
 64 bytes from 10.180.0.2: icmp_seq=3 ttl=64 time=9.22 ms
 64 bytes from 10.180.0.2: icmp_seq=4 ttl=64 time=9.28 ms
 64 bytes from 10.180.0.2: icmp_seq=5 ttl=64 time=9.23 ms
 64 bytes from 10.180.0.2: icmp_seq=6 ttl=64 time=9.27 ms
 64 bytes from 10.180.0.2: icmp_seq=7 ttl=64 time=9.24 ms
 64 bytes from 10.180.0.2: icmp_seq=8 ttl=64 time=9.22 ms
 64 bytes from 10.180.0.2: icmp_seq=9 ttl=64 time=9.24 ms
 64 bytes from 10.180.0.2: icmp_seq=10 ttl=64 time=9.22 ms
 64 bytes from 10.180.0.2: icmp_seq=11 ttl=64 time=9.23 ms
 64 bytes from 10.180.0.2: icmp_seq=12 ttl=64 time=9.22 ms
^C

```

- From the figure in the previous step. What facilitates this connectivity: the virtual switch or the VPN Gateway?

The virtual switch! The VPN gateway doesn't really have scope of the other subnets. For the virtual switch, the other subnets are kind of like hosts on a network. They are within its scope and reachable.

```

jgafron@cloudshell:~ (cloud-gafron-jgafron)$ gcloud compute networks subnets list --regions=us-central1,europe-west1
NAME: default
REGION: europe-west1
NETWORK: default
RANGE: 10.132.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:

NAME: subnet-europe-west-192
REGION: europe-west1
NETWORK: custom-network1
RANGE: 192.168.5.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:

NAME: default
REGION: us-central1
NETWORK: default
RANGE: 10.128.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:

NAME: subnet-us-central-192
REGION: us-central1
NETWORK: custom-network1
RANGE: 192.168.1.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:

```

- Explain why the result of this ping is different from when you performed the ping to **instance-2**.

Because Instance-1, and the newly created instances are on completely different networks! The instances we created are also mostly likely protected by google cloud VMs infrastructure, and therefore not open to connections from the outside internet. In order to successfully ping the newly created instances, we would need to configure or create the routes in order to do so.

```
jgafron@instance-1:~$ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
^C
--- 192.168.1.2 ping statistics ---
71 packets transmitted, 0 received, 100% packet loss, time 71663ms

jgafron@instance-1:~$ ping 192.168.5.2
PING 192.168.5.2 (192.168.5.2) 56(84) bytes of data.
^C
--- 192.168.5.2 ping statistics ---
11 packets transmitted, 0 received, 100% packet loss, time 10241ms
```

- Take screenshots of all 4 instances in the UI including the network they belong to.

Filter Enter property name or value										
☐	Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Network	Connect	
☐	✓	course-vm	us-west1-b			10.138.0.2 (nic0)	34.169.132.94 (nic0)	default	SSH	⌵ ⋮
☐	✓	instance-1	us-west4-b			10.182.0.2 (nic0)	34.125.120.28 (nic0)	default	SSH	⌵ ⋮
☐	✓	instance-2	us-west3-b			10.180.0.2 (nic0)	34.106.101.90 (nic0)	default	SSH	⌵ ⋮
☐	✓	instance-3	us-central1-a			192.168.1.2 (nic0)	34.42.178.20 (nic0)	custom-network1	SSH	⌵ ⋮
☐	✓	instance-4	europe-west1-d			192.168.5.2 (nic0)	34.78.120.168 (nic0)	custom-network1	SSH	⌵ ⋮

Sort Enter property name or value									
Name ↑	Subnets	MTU ⓘ	Mode	IPv6 ULA range	Gateways	Firewall rules	Global dynamic routing	Network profile	
custom-network1	2	1460	Custom			0	Off		
default	41	1460	Auto			4	Off		

custom-network1

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OVERVIEW

SUBNETS

STATIC INTERNAL IP ADDRESSES

FIREWALLS

FIREWALL ENDPOINTS

ROUTES

VPC NETWORK PEERING

PRIVATE

>

Subnets

+

ADD SUBNET

≡

MANAGE FLOW LOGS

Filter

Enter property name or value

?

≡

Name ↑	Region	Stack Type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal ranges	Gateway	
subnet-europe-west-192	europe-west1	IPv4 (single-stack)	192.168.5.0/24			None	192.168.5.1	▼
subnet-us-central-192	us-central1	IPv4 (single-stack)	192.168.1.0/24			None	192.168.1.1	▼

default

<

OVERVIEW

SUBNETS

STATIC INTERNAL IP ADDRESSES

FIREWALLS

FIREWALL ENDPOINTS

ROUTES

VPC NETWORK PEERING

Subnets

+

ADD SUBNET

≡

MANAGE FLOW LOGS

Filter

Enter property name or value

?

Name ↑	Region	Stack Type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal ranges	Gateway	
default	africa-south1	IPv4 (single-stack)	10.218.0.0/20			None	10.218.0.1	
default	asia-east1	IPv4 (single-stack)	10.140.0.0/20			None	10.140.0.1	
default	asia-east2	IPv4 (single-stack)	10.170.0.0/20			None	10.170.0.1	
default	asia-northeast1	IPv4 (single-stack)	10.146.0.0/20			None	10.146.0.1	
default	asia-northeast2	IPv4 (single-stack)	10.174.0.0/20			None	10.174.0.1	
default	asia-northeast3	IPv4 (single-stack)	10.178.0.0/20			None	10.178.0.1	